Fast-Wave, High-Efficiency, Wideband, Meta-Line Antenna Design Based on Metal Bezel for 5G Smartphones

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***Abstract*—The design of a novel meta-line antenna (MLA) from a normal inverted L antenna is presented. By adding series capacitors within the bezel slit and shunt inductors to ground, a fast-wave electrically large antenna is obtained. With capacitive and inductive loadings, an obvious upshifting resonant frequency is observed, which gives an opportunity for the application of an antenna with longer dimensions. It is shown as an obvious enhancement of antenna efficiency. In addition, a wideband, tunable MLA design for the middle-to-high frequency band is presented, leveraging the metal bezel of a smartphone. The prototype is fabricated and measured, and the results are compared to conventional designs, which indicate potential applications for cellular communications, particularly during voice calls.**

***Index Terms*—Fast-wave antenna, meta-line antenna (MLA), mobile antenna, small antenna, terminal antenna.**

I. Introduction

HIGH-EFFICIENCY and wide-bandwidth antennas are required in various kinds of modern terminal products, including smartphones, smartwatches, true wireless stereo earphones, and tablet personal computers. To achieve aesthetic industrial design, metal bezels, especially those surrounding the screen, are commonly integrated as antennas in mainstream products, often serving as the devices' skeleton. Inverted L/F antenna, dipole antenna, and slot antenna are widely utilized based on these metal bezels [1], [2], [3], [4], [5], [6], [7], However, as products strive to become thinner and lighter, structural constraints are pushing design limits, resulting in reduced antenna clearance. This reduction significantly degrades antenna efficiency and bandwidth.

To overcome these challenges, extensive research has focused on enhancing antenna efficiency and bandwidth. A typical design involved using parasitic resonant loadings. Metal strip [8], [9], split-ring resonators [10], and loops [11] are designed to improve the performance of wire antennas. In particular, Liu et al. [12] examined how parasitic structures influence antenna efficiency and the underlying mechanisms. Guided by this principle, Li et al. [13] introduced a parasitic structure that achieves in-phase current to enhance efficiency. However, each of these works requires multiple structures, resulting unavoidably in either complicated geometrical configurations or considerable size parameters. Alternatively, the mode compression method offers an effective solution for bandwidth expansion in physically constrained antennas. Typically, multiple resonant modes are excited, with their frequencies carefully designed to be closely spaced through the use of loaded structures. Similar methods have demonstrated comparable effectiveness when applied to dipole [14], [15], slot [16], [17], and patch antennas [18], [19]. While this method enhances impedance bandwidth, it fails to bring about improvement in efficiency.

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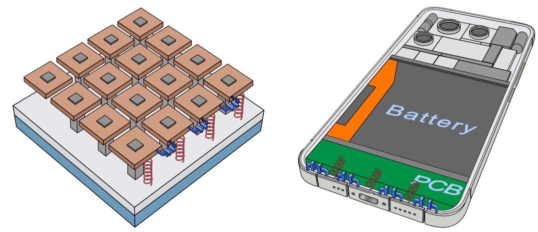

(a) (b)

Fig. 1. Illustration of conventional (a) meta surface and (b) meta-line structure, which is used for antennas applied to 5G smartphone.

Since the inception of the mushroom-like structure [20], [21], research on meta-surface antennas has gained tremendous momentum over the past decade. In [22] and [23], a close combination between TM_{01} and antiphase TM_{02} modes in periodic patches was achieved, which obtained a broad impedance bandwidth with stable boresight radiation patterns. Wei et al. [24] presented a loop antenna based on mu-negative transmission line, achieving a wide bandwidth and a thin structure. In [25], an interdigital structure was employed to create distributed capacitors, and a wideband dipole was achieved. Notably, all of these structures provide negligible enhancement to the antenna's radiation efficiency.

Inspired by meta-surface, a 1-D meta-line antenna (MLA), as drawn in Fig. 1, is first presented for smartphones with a significant efficiency enhancement. The rest of this letter is organized as follows. In Section II, we first modify a conventional "T" antenna into a meta-line structure, and then propose a practical cellular antenna design for smartphones based on a metal bezel. We discuss simulated, measured, and compared results in Section III. Finally, Section IV concludes this letter.

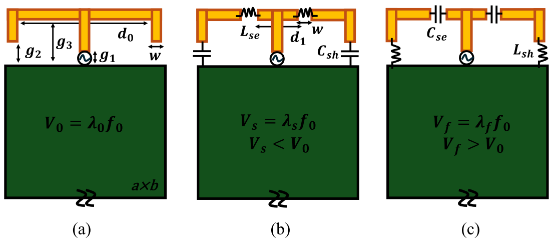

Fig. 2. "T" antennas loading with *LC* passive elements. (a) Conventional T antenna. (b) Slow-wave antenna with series inductors and shunt capacitors. (c) Fast-wave antenna with series capacitors and shunt inductors. Note that the subscript "s" denotes "slow," and "f" denotes "fast."

TABLE I
DETAILED PARAMETERS

Parameter	Value (mm)	Parameter	Value (mm)	Parameter	Value
d_0	32	g_3	5	L_{se}	3.3 nH
d_1	10	w	2	L_{sh}	12 nH
g_1	1	a	150	C_{se}	0.2 pF
g_2	2	b	70	C_{sh}	0.5 pF

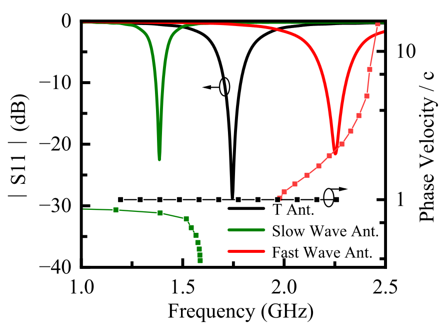

Fig. 3. Reflection coefficients and phase velocities of the "T" antennas with different loaded capacitors and inductors.

II. Antenna Design and Operating Principles

A. *Concept of Fast-Wave MLA*

A typical T-shaped antenna is designed above the printed circuit board (PCB). The geometry structure of the proposed antenna is shown in Fig. 2. The dipoles are bent and with a typical clearance of 1 mm, as depicted in Fig. 2(a). A central fed antenna structure is adopted, and a simple shunt matching network is incorporated to ensure 50 Ω input impedance. The clearance is filled with FR4 substrate with dielectric constant of 4.3 and a loss tangent of 0.025. In Fig. 2(b) and (c), the antennas share an identical configuration, yet are engineered with symmetric slots to achieve meta-line structure. The fracture position is strategically located adjacent to the feeding points where it is close to a strong current, and the detailed dimensions are listed in Table I.

Resonance study and dispersion relationship were performed using T-solver and E-solver of CST, respectively. As drawn in Fig. 3, the original "T" antenna resonates at band 3 (B3) of 5G new radio (NR).

By introducing the fracture, the weakened coupling between gaps results in radiation primarily from the middle short T-antenna, shifting its resonant frequency beyond the target range. The inductors in Fig. 2(b) are strategically positioned in

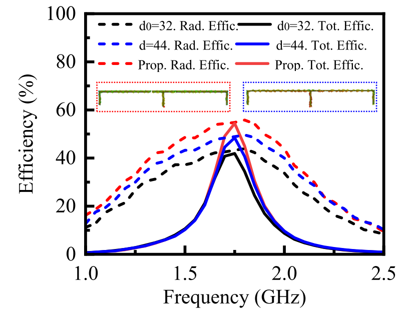

Fig. 4. Efficiencies of the original "T" antenna, enlarged reference antenna, and proposed fast-wave MLA.

regions of high current density, whereas the capacitors are placed in areas of strong electric field intensity. As a result, the extended electrical length leads to a decrease in the resonant frequency, as shown in Fig. 3. Again, by replacing the original components with series capacitors and shunt inductors in Fig. 2(c), the resonant frequency is increased. Although the series capacitor enhances coupling, it is still weaker than the metal-to-metal connection without any gap, resulting in the reduced electrical length.

Phase velocities are normalized to the speed of light and plotted in Fig. 3. The slow-wave antenna exhibits a phase velocity $V_s < c$, and a wavelength $\lambda_s < \lambda_{0,}$ enabling antenna size reduction. Conversely, the fast-wave antenna, characterized by $V_s > c$ and $\lambda_s > \lambda_0$, leads to an enlarged antenna size. We enlarge the floating arm of the fast-wave structure, maintaining an approximate 2:1 ratio between the slot-to-end and slot-to-feed distances. The initial dimension of $d_0 = 32$ mm is enlarged to $d_f = 44$ mm, thereby tuning the resonant frequency to B3.

A d = 44 mm reference antenna without slots is also tuned to B3 for comparison. As illustrated in Fig. 4, the fast-wave antenna exhibits a 11% efficiency improvement over the original "T." Enhancement can be attributed to: 1) an enlarged aperture, which directly boosts efficiency and 2) more uniform, codirectional currents facilitated by the slot structure. A uniform current distribution across the fast-wave structure, exhibiting lower peak and average current densities, directly translates to reduced metal losses.

B. *Cellular Antenna Design*

A cellular communication antenna is designed at the bottom of a smartphone. Fig. 5 shows the detailed configuration of the proposed antenna. The antenna employs three bezel segments, featuring two grounded side sections, and a central floating stub, which is commonly used for USB to transmit and receive data or charge battery. On the one side of the floating stub is the feeding together with a shunt tuner to ensure the cellular antenna covering the total middle-to-high frequency band (MHB) of 1.71 GHz to 2.69 GHz. On the other side of the stub, tunable series capacitors and shunt inductors are used to realize the meta-line structure. Next to the feeding, another parasitic inverted-L stub with shunt switch is also used to broaden the bandwidths.

In this design, two types of switches are required: 1) single-pole single-throw to connect passive elements between PCB ground and antenna and 2) series switch (e.g., SP4T) to connect

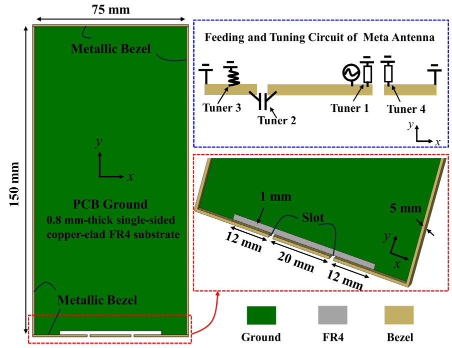

Fig. 5. Geometry of the fast-wave MLA based on metallic bezel. The “red dotted” rectangular indicates the detail configuration of the antenna at the bottom of smartphone, and the “blue dotted” rectangular shows the feeding and tuning circuit of the antenna.

TABLE II
DETAILED PARAMETERS

Band \ Value \ Switch	Tuner1	Tuner2	Tuner3	Tuner4
B3	OFF	0.5 pF	OFF	0.4 pF
B1	27 nH	0.5 pF	OFF	0.2 pF
B40	18 nH	0.4 pF	6.8 nH	OFF
B7/B41	5.6 nH	0.4 pF	2.7 nH	33 nH

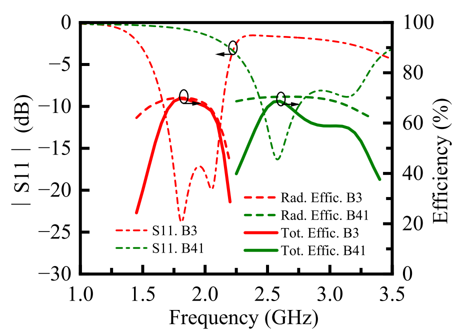

Fig. 6. Simulated *S*-parameters and efficiencies of MLA with parasitic monopole structure.

components between antenna branches. The component values for four typical bands are listed in Table II. Fig. 6 presents the reflection coefficients and efficiencies, showing only the first B3 and last B41 for conciseness. Within each operating band, the peak efficiency can reach 70%. Importantly, low-conductivity copper was employed in simulation to accurately represent the significant loss contribution of the metal branch under real-world conditions.

Fig. 7 depicts the current distributions at both resonant frequencies. At 1.8 GHz, the dominant radiation originates from the left meta structures, characterized by intense, same-directional currents. In contrast, the right parasitic monopole exhibits synchronized but comparatively weak currents. The same-directional current flow across the three separated stubs contributes to high radiation efficiency. At 2.05 GHz, the roles shift, the right parasitic monopole serves as the primary radiating branch, and opposite directional current emerges, leading to a gradual decay in efficiency. Around 2.25 GHz, the currents in meta-stubs and monopole-stubs exhibit completely out-of-phase behavior across all phases. A significant dip in efficiency is observed.

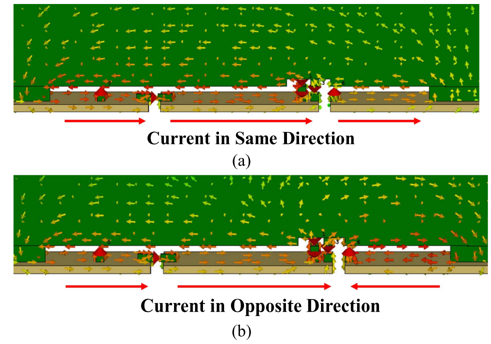

Fig. 7. Current distributions of meta-antenna with parasitic monopole at frequency of B3. (a) 1.8 GHz. (b) 2.05 GHz.

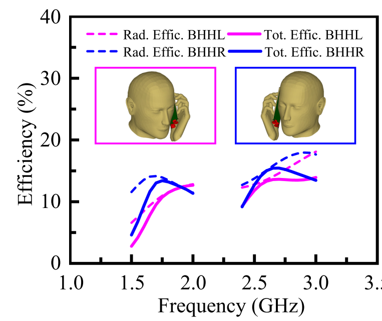

Fig. 8. Simulated efficiencies of MLA beside head and hand in B3 and B41, respectively.

The bottom-mounted antenna is designed to remain distant from both the hand and the head, ensuring optimal performance in scenarios where the user holds the phone to their head for calls. The proposed MLA is simulated beside the head and hand left-side (BHHL), and right-side (BHHR), respectively. These phantoms adhere to CTIA standards. The resulting simulated efficiencies are shown in Fig. 8. The resonances of BHHL shift by about 40 MHz. Nonetheless, an average BHHL and BHHR efficiency of 13.6% is obtained in B3, and 14.1% in B41. When compared to free-space performance, the BHH efficiency exhibits a degradation of about 56%, which is competitive for modern antenna design.

III. Measurement and Comparison

To validate the design, a parasitic inverted L antenna (ILA)-loaded MLA was fabricated and measured. Furthermore, an ILA and a fast-wave MLA were implemented for direct comparison as control groups. Fig. 9 shows the 2-D configuration of each antenna. We employed a copper alloy with low electrical conductivity to more accurately represent the efficiency of real antennas in metal-bezel smartphones. Fig. 10(a) gives a photograph of the fabricated samples. The bezel is welded to the ground with the help of conductive adhesive, and clearances are produced by a laser cutting machine. Semirigid cables with SMA connectors are used for feedings.

The antennas are measured using an Agilent E5071C vector network analyzer. For clarity in visualization, Fig. 10(b) only presents the results of ILA and MLA in B3. As seen, the curves of ILA and MLA fluctuate in the high-frequency band, and the curves of MLA with Para. exhibit sharp minor fluctuations. They originate from semirigid cable, as the exceptional point exhibits

TABLE III
COMPARISON OF THE METAL-BEZEL ANTENNAS IN SMARTPHONE

Ref.	Electrical conductivity	Bezel windowing	Electrical size (λ_0)	Impedance bandwidth (GHz)	Average total efficiency (%)	Peak efficiency (%)	Antenna type
[1]	General	Yes	0.550	1.71–2.69	59	67	Slit+Loop
[2]	General	No	0.375	1.71–2.69	54	62	IFA+Para.
[3]	General	No	0.432	1.71–1.92, 2.3–2.5	58.8, 56.9	89.1, 86.3	T+Dipole
[6]	General	Yes	0.880	3.4–4.2	55.3	67.5	Dipole
[7]	General	No	0.402	1.71–2.99	54.6	74	IFA+Para.
[13]	General	No	0.39	0.85–0.95	39.8	41.6	IFA+Para.
Control-1	Low	No	0.1	1.71–1.88	37.2	43.65	ILA.
Control-2	Low	No	0.2	1.6–2.2	43.6	57.5	MLA.
Proposed	Low	No	0.28	1.6–2.2, 2.2–3.25	51.3, 49	64.6, 63.1	MLA+Para.

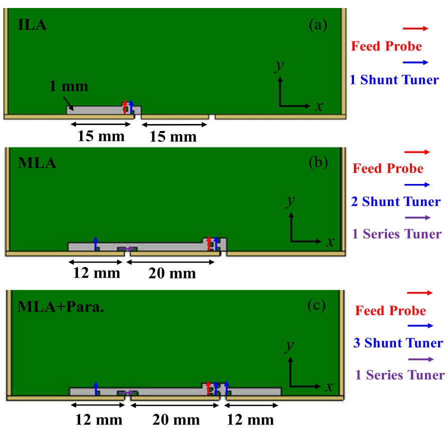

Fig. 9. 2-D geometry of (a) normal ILA, (b) MLA, and (c) MLA loaded with parasitic ILA.

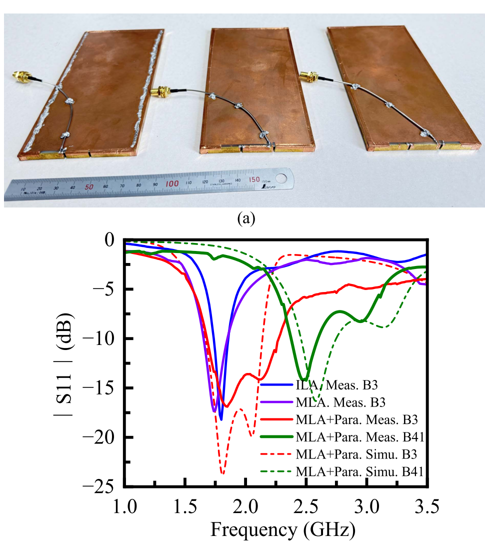

Fig. 10. Antenna measurements. (a) Photograph of the fabricated ILA, MLA, and MLA loaded with parasitic ILA. (b) Simulated and measured S-parameters.

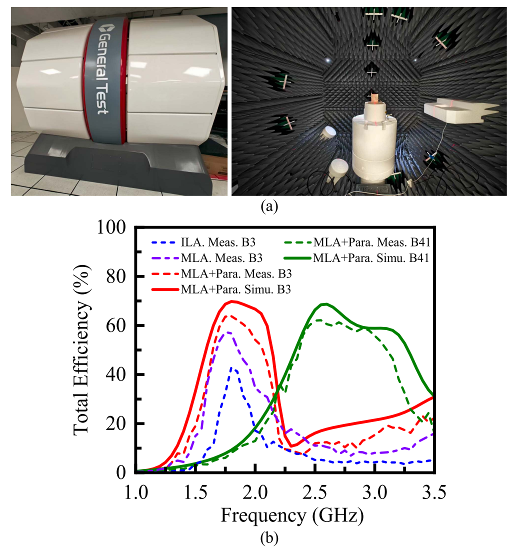

Fig. 11. Efficiency measurements. (a) Scenario in chamber. (b) Simulated and measured efficiencies.

significant sensitivity to manual contact. In the fabrication process, it is essential that the cables are securely fastened and adequately grounded. For the MLA with parasitic ILA, the measured results for B3 and B41 shift up 2.1% and shift down 4.6% with respect to the simulated results. Quantitative sensitivity analysis reveals that slot width and capacitor inaccuracies can affect results.

The antenna efficiencies were measured using the RayZone 2800 system from General Test, Inc., in an anechoic chamber. The photograph of the measurement scenario is given in Fig. 11(a), and the results are plotted in Fig. 11(b). Efficiencies of MLA with Para. are measured, for B3 and B41, respectively. The measured results generally agree well with simulations, slight frequency shifts are noted, and the observed loss of less than 10% loss is attributed to the cables. The proposed antenna is further compared to both ILA and MLA in B3. As seen, the MLA exhibits a higher peak value and a wider bandwidth compared to the traditional ILA. Furthermore, the proposed antenna achieves its enhanced bandwidth by incorporating parasites. Table III compares the measured antennas with state-of-the-art antennas from references.

IV. CONCLUSION

In this letter, a novel fast-wave electrically large MLA is designed using the metal bezel of smartphones. To enhance the efficiency bandwidth, a parasitic ILA is also incorporated. This antenna is engineered for high-efficiency performance, especially in the face of ever-shrinking clearances. The fast-wave meta-line structure has been substantiated through the use of lumped capacitors and inductors, and the MLA with parasite has been fabricated and measured to cover all the MHB of 5G NR with the help of tuners. Beyond that, a comparative study with normal ILA and MLA is conducted. It is shown that the MLA outperforms normal ILA with better impedance bandwidth and higher efficiency, although at the expense of physical size.

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